Solutions and teaching notes for chapter 10

Teacher use. The answers below concern the tasks not worked in the lesson. First have students state their method; give graded help before the solution. The durations are estimates to be calibrated in class.

Diagnostic and remediation

1. 2π. *Action:* Link radius, diameter and perimeter.
2. $90°$. *Action:* Divide a circle into four sectors.
3. $\dfrac{\sqrt{2}}{2}$. *Action:* Rationalise and check the square.
4. $(a, -b)$. *Action:* Draw the two points.
5. $-\dfrac{\pi}{3}$. *Action:* Put over a common denominator.
6. $\dfrac{3}{5}$. *Action:* Identify the reference angle.

Investigation activity

Equilateral triangle then isosceles right triangle. Pythagoras gives respectively height $\dfrac{\sqrt{3}}{2}$ and equal coordinates $\dfrac{\sqrt{2}}{2}$. The signs are positive in the first quadrant.

Immediate applications

1. $\dfrac{\pi}{4}$, $\dfrac{2\pi}{3}$; length $\dfrac{3\pi}{2}$ m.
2. $\dfrac{7\pi}{2}$ has the image of $\dfrac{3\pi}{2}$; $-\dfrac{5\pi}{3}$ that of $\dfrac{\pi}{3}$.
3. $\sin^2 x = \dfrac{16}{25}$; the sine is negative, so $-\dfrac{4}{5}$.
4. $(-\dfrac{1}{2}, -\dfrac{\sqrt{3}}{2})$ and $(\dfrac{\sqrt{2}}{2}, -\dfrac{\sqrt{2}}{2})$.
5. $h = \dfrac{\sqrt{3}}{2}$; new chord $\dfrac{1}{\sqrt{2+\sqrt{3}}} = \sqrt{2-\sqrt{3}}$.

Exercises: method and results

Exercise 10.1 – Converting and measuring. $\dfrac{5\pi}{6}$ and $-\dfrac{\pi}{2}$; arc $\dfrac{4\pi}{3}$ cm. The negative sign indicates an orientation; a geometric length is positive.

Exercise 10.2 – Reducing angles. $\dfrac{5\pi}{6}$, $\dfrac{\pi}{4}$, π. Remove or add multiples of 2π.

Exercise 10.3 – Exact values. $(-\frac{1}{2}, \frac{\sqrt{3}}{2})$; $(\frac{\sqrt{3}}{2}, -\frac{1}{2})$; $(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2})$, in the form (cosine; sine).

Exercise 10.4 – Recovering the sign. $\cos^2 x = 1 - \frac{25}{169} = \frac{144}{169}$; cosine negative, so $-\frac{12}{13}$.

Exercise 10.5 – A wheel. Advance $r\theta = 0.3(\frac{5\pi}{3}) = \frac{\pi}{2}$ m. This corresponds to the unrolling of the arc; the chord joins two positions on the circle and does not have this length. A material point on the wheel follows a different path relative to the ground.

Exercise 10.6 – Chord and scalar product. $A(1,0)$, $B(-\frac{1}{2}, \frac{\sqrt{3}}{2})$; $AB^2 = \frac{9}{4} + \frac{3}{4} = 3$. Al-Kashi: $1 + 1 - 2\cos(\frac{2\pi}{3}) = 3$. So $AB = \sqrt{3}$.

Exercise 10.7 – Reading an equation on the circle. The vertical line with x-coordinate $\frac{1}{2}$ meets the circle at the angles $\frac{\pi}{3}$ and $\frac{5\pi}{3}$. The two points have opposite sines.

Exercise 10.8 – Delayed review. $\frac{25\pi}{4} = 6\pi + \frac{\pi}{4}$. Three complete turns do not change the image, so both values are $\frac{\sqrt{2}}{2}$.

CQFD, exercise 32 p. 220, questions a and b. a: $\cos(\frac{6\pi}{5}) = \cos(\pi + \frac{\pi}{5}) = -\frac{1+\sqrt{5}}{4}$. b: $\cos(-\frac{\pi}{5}) = \cos(\frac{\pi}{5}) = \frac{1+\sqrt{5}}{4}$, by evenness of the cosine.

CQFD, exercise 39 p. 220, questions 1, 2, 3 and 5. 1: $\frac{\pi}{6} + \frac{\pi}{3} = \frac{\pi}{2}$. 2: yes, $\cos(\frac{\pi}{3}) = \frac{1}{2} = 0.5$ is a decimal. 3: the principal measure in $(-\pi, \pi]$ is $-\frac{5\pi}{6}$. 5: $\sin(\frac{4\pi}{3}) = -\frac{\sqrt{3}}{2}$.

MCQ: answers and reasons

MCQ 1.

1. B. Half turn.
2. A. Equilateral triangle.
3. D. Lowest point of the circle.
4. C. Equation of the unit circle.
5. B. Complete turns.

MCQ 2.

1. C. Vertical symmetry.

2. A. Isosceles right triangle.
3. D. Three quarters of a turn.
4. B. $r\theta = 6$.
5. C. Negative x-coordinate inside the quadrant.

True or false

1. False: at 0, the two sides would be 1 and -1. The cosine is even.
2. True: a complete turn keeps the image.
3. False: $\frac{\pi}{6}$ and $\frac{5\pi}{6}$ have the same sine but opposite cosines.

Differentiated teaching. For a technical block, offer only the first step (domain, factor or reference sample space), then withdraw the help. For an independent student, ask for a second method and the treatment of a limiting case. Orally, allow an initial reformulation in the language the student has mastered, then require a precise mathematical sentence in French.